(1) 12. Use a multi-step procedure leading to a mathematical solution

Solving must follow clear steps:

1. Understand the regulation: “Minimum distance between towers = 5 × rotor blade length.”
2. Extract rotor blade length: given as 40 meters.
3. Calculate the minimum allowed distance: $5 \times 40 = 200$ meters.
4. Analyze the tower layout in the diagram (given a 500×500 m field with towers arranged in a grid).
5. Measure or calculate the actual distance between adjacent towers in the diagram.
6. Compare the actual distance with the minimum allowed distance.
7. Draw a conclusion: state whether the layout complies and explain why.

(2) 6. Extract mathematical information from diagrams, graphs, or constructions

- The question provides a **layout diagram** with towers represented as dots.
- Students need to **infer tower coordinates or relative positions** from the diagram and combine this with the field dimensions (500 m) to calculate actual distances.
- They must understand the meaning of “drawing not to scale” but still perform **spatial reasoning** based on the grid layout.

(3) 14. Make generalisations based on the results of applying mathematical procedures

- After calculating the minimum allowed distance (200 m), it must be compared with the actual layout distances.
- If any adjacent tower spacing is found to be < 200 m, students must **judge** that the layout does not comply.
- They must support their argument with calculations and provide a **general conclusion** (e.g., “The mayor’s suggestion violates regulations because distance X meters < 200 meters”).

(4) 7. Manipulate numbers, algebraic expressions, and geometric representations

- Numerical calculation: $5 \times 40 = 200$.
- Geometric reasoning: calculate distances between points in a square grid (possibly using the Pythagorean theorem or grid-counting methods).
- Understanding of scale and space: map the diagram layout to real-world dimensions (e.g., what each grid unit corresponds to in meters).

(5) 8. Articulate a solution, showing intermediate mathematical results

- Students must **explain their reasoning in writing**.
- They should show calculation steps (minimum distance calculation, layout distance estimation).
- The final conclusion must be clear and **supported by calculated data**.

(6) 10. Use a variety of representations when interacting with a problem

- Textual description (regulation explanation).
- Numerical parameters (blade length, field size).
- Schematic diagram (spatial layout).
- Algebraic/geometric model (distance formula).

（1）12. 使用多步程序得出数学解、结论或概括

解题必须遵循明确的步骤：

1. 理解规范："塔间最小距离 = 5 × 转子叶片长度"。
2. 提取叶片长度：已知为 40 米。
3. 计算最小允许距离：$5 \times 40 = 200$ 米。
4. 分析示意图中塔的布局（已知场地为 500×500 米，塔按网格排列）。
5. 测量或计算图中相邻塔的实际距离。
6. 比较实际距离与最小允许距离。
7. 得出结论：说明是否合规，并解释理由。

（2）6. 从数学图表、图形、构造或计算工具中提取数学信息

- 题目提供了一张**布局示意图**，塔的位置用点表示。
- 学生需要从图中**推断塔的坐标或相对位置**，并结合场地尺寸（500 米）推算实际距离。
- 必须理解"图不按比例绘制"的含义，但仍需根据网格布局进行**空间推理**。

（3）14. 基于运用数学程序得出的结果进行概括或判断

- 计算出最小允许距离（200 米）后，需与实际布局距离比较。
- 如果发现任一相邻塔间距 < 200 米，则需**判断**该布局不合规。
- 需用计算结果支持论证，并给出**概括性结论**（如"市长建议违反规定，因为距离 X 米 < 200 米"）。

（4）7. 操作数字、图表和统计信息、代数表达式和几何表示

- 数值计算：$5 \times 40 = 200$。
- 几何推理：在正方形网格中计算点之间的距离（可能用到勾股定理或网格数法）。
- 空间与比例理解：将图示布局映射到实际尺寸（如每个网格单位对应多少米）。

（5）8. 清晰表达解题过程，展示或总结中间数学结果

- 学生需**书面解释**推理过程。
- 应展示计算步骤（最小距离计算、布局距离推算）。
- 最终结论需明确，并**引用计算数据**作为支持。

（6）10. 在问题解决中使用多种表征形式

- 文字表述（规定说明）。
- 数值参数（叶片长度、场地尺寸）。
- 示意图（空间布局）。
- 代数/几何模型（距离公式）。